

MASTER'S COMPREHENSIVE EXAMINATION

Part I: Theory (90 minutes) April 2, 2011

Instructions: Attempt any three out of the following four questions. Mention which three you would like to have graded. Open book and open notes.

1. X_1, X_2, X_3 are Bernoulli variables with parameter p .

a) Test $H_0: p = 1/4$ vs $p = 1/2$ at level .0156. That is, determine explicitly the rejection region of the most powerful test.

b) Find the power of the test in a).

2. Let X_1, X_2, X_3 be three independent normal random variable with means (μ_1, μ_2, μ_3) respectively and variances $(\sigma_1^2, \sigma_2^2, \sigma_3^2)$, respectively.

a) What is $P\{X_1 + X_2 + X_3 > 4\}$.

b) What is the variance of $Y = X_1 X_2$.

3. A box contains 4 red balls and 6 black balls. Three balls are picked at random (without replacement) from the box, and X , the “number of red balls” is observed. If $X = k$, then k six sided fair dice are rolled, and Y , the “combined total number of spots on the dice is recorded. (If $X = 0$, then $Y = 0$).

a) Find the expectation of $\{Y\}$ _____

b) Find $\text{Prob}\{Y=3\}$ _____

4. Let the joint probability density function of the continuous random variables X and Y be

$$f(x,y) = C \exp\{2x + 3y\} \quad \text{for } 0 < x < y, \quad 0 < y < 1$$

$$= 0 \quad \text{elsewhere}$$

a) Find C _____

b) Find $P(Y < 0.3)$ _____
